

Impact of AI on Society and Employment

- **Profound Reshaping:** The advent of artificial intelligence has profoundly reshaped the fabric of society and employment.
- **Workforce Impact:** AI's influence alters job landscapes, raising ethical concerns about job displacement.
- **Ethical Governance:** There is an urgent need for comprehensive strategies to govern AI's ethical implications.
- **Sector Influence:** AI permeates various sectors, from healthcare to finance, affecting societal norms.

AI's Impact on Employment

- **Job Displacement Risk:** Approximately 47% of jobs in the United States are at risk of automation according to a study by Frey and Osborne.
- **Necessity for Actionable Insights:** Professionals need to develop frameworks to mitigate the adverse impacts of AI on the workforce.
- **Job Redesign Strategies:** Organizations can implement strategies emphasizing the augmentation rather than replacement of human labor.
- **Hybrid Roles Creation:** Businesses can create hybrid roles that combine human creativity and emotional intelligence with machine efficiency.

Human-Machine Teaming Framework Overview

- Encouraging Collaboration: The Human-Machine Teaming Framework encourages collaboration between humans and AI systems.
- Task Identification: This framework guides organizations in identifying tasks suitable for automation while retaining human oversight.
- Industry Application: Industries like healthcare have successfully applied this approach, where AI aids in diagnostics but human doctors deliver patient care.
- Organizational Benefits: By adopting such frameworks, organizations can not only preserve jobs but also enhance productivity and innovation.

AI's Societal Impact

- Influence on Social Structures: AI's societal impact extends beyond employment, influencing social structures and ethical norms.
- Privacy Concerns: The proliferation of AI technologies raises privacy concerns, as data collection becomes more pervasive.
- Role of AI Ethics Professionals: Professionals in AI ethics must address these concerns by implementing robust data governance frameworks.
- Data Ethics Framework: The Data Ethics Framework, developed by the UK government, provides a comprehensive guide for ethical data management.

Ethical Management of AI and Societal Inequalities

- **Exacerbating Inequalities:** AI has the potential to exacerbate societal inequalities if not managed ethically, leading to discriminatory outcomes.
- **Algorithmic Bias:** Algorithmic bias can stem from training data that reflects existing societal prejudices.
- **Hiring Process Impact:** AI systems used in hiring processes may inadvertently favor certain demographic groups, perpetuating inequality.
- **Counteracting Bias:** Utilize the Fairness, Accountability, and Transparency in Machine Learning framework to identify and mitigate bias in AI systems.

Societal Implications of AI

- **Accountability and Decision-Making:** AI systems challenge traditional notions of responsibility, raising questions about liability in cases of malfunction or harm.
- **Human Oversight:** The EU's Ethics Guidelines emphasize the importance of human oversight and accountability in AI systems.
- **Navigating AI Ethics:** By adhering to these guidelines, professionals can ensure AI systems align with societal values and ethical standards.

AI Integration in Banking: Case Study Insights

- **Practical Application:** Case studies illustrate the practical application of these tools and frameworks, offering valuable insights for professionals.
- **Fraud Detection Benefits:** Banks using AI for fraud detection have significantly reduced fraudulent activities, showcasing AI's ability to enhance security and efficiency.
- **Challenges of Bias:** The same systems have faced criticism for algorithmic bias, highlighting the need for robust ethical frameworks to ensure fairness and accountability.

The Importance of Ethical AI Governance

- **Economic Potential:** AI adoption could contribute an additional \$13 trillion to global GDP by 2030, highlighting its significant economic potential.
- **Ethical Concerns:** This growth relies on addressing ethical concerns and ensuring AI benefits are equitably distributed.
- **Inclusive Policies:** Professionals must advocate for inclusive AI development to bridge the digital divide and prevent exacerbating existing inequalities.

Ethical Governance in AI Integration

- **Proactive Approach:** The integration of AI into society necessitates a proactive approach to ethical governance.
- **Actionable Insights:** Professionals equipped with actionable insights and strategies can effectively manage the impact of AI on employment and society.
- **Multidisciplinary Perspective:** By embracing a multidisciplinary perspective that combines technical expertise with ethical considerations, professionals can lead the charge in shaping AI governance.

UNESCO(United Nations Educational Scientific and Cultural Orgnaisation)

- Adopted by 123 states in November 2021.

Core Values

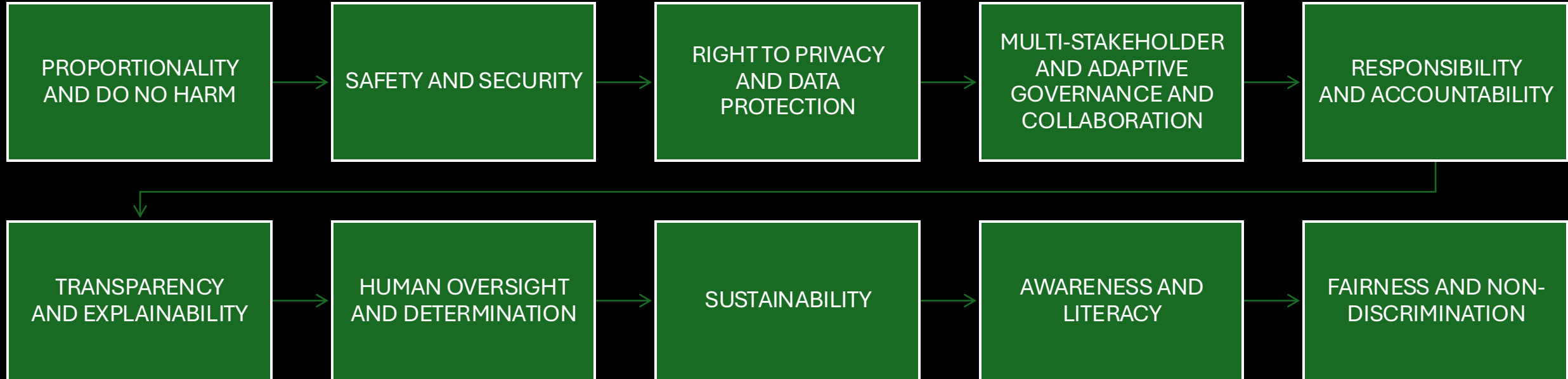
RESPECT, PROTECTION AND PROMOTION OF
HUMAN RIGHTS AND FUNDAMENTAL FREEDOMS
AND HUMAN DIGNITY.

LIVING IN PEACEFUL, JUST, AND
INTERCONNECTED SOCIETIES.

ENSURING DIVERSITY AND INCLUSIVENESS.

ENVIRONMENT AND ECOSYSTEM FLOURISHING

10 core principles



1) Proportionality and Do No Harm

- AI systems should be designed to benefit society and avoid causing physical, psychological, or social harm.
- The level of AI intervention must be proportional to the real need and risk involved.
- Risk assessment and impact analysis should be conducted before deployment.
- Developers must prevent misuse such as surveillance abuse, misinformation, or harmful automation.
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2) Safety and Security

- AI systems must be reliable, robust, and resistant to cyber-attacks.
- Continuous testing and validation should ensure system stability in real-world environments.
- Secure data pipelines and infrastructure must be maintained.
- Emergency fallback mechanisms should be available if AI fails.

3) Right to Privacy and Data Protection

- AI must respect personal data rights and follow data protection laws.
- Users should know what data is collected and how it is used.
- Data should be minimized, anonymized, and securely stored.
- Individuals must have control over their personal data and consent.

4) Multi-Stakeholder and Adaptive Governance & Collaboration

- AI governance should involve governments, industry, academia, and society.
- Policies must adapt as technology evolves.
- International cooperation is essential for global AI regulation.
- Inclusive participation ensures diverse perspectives in decision-making.

5) Responsibility and Accountability

- Developers and organizations must be accountable for AI outcomes.
- Clear liability frameworks should exist for AI failures or harm.
- Ethical guidelines must be embedded in the development lifecycle.
- Regular audits and compliance checks

6) Transparency and Explainability

- AI decisions should be understandable and interpretable.
- Users must know when they are interacting with AI systems.
- Black-box decision making should be minimized.
- Explainability builds trust and enables error correction.
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7) Human Oversight and Determination

- Humans must remain in control of critical AI decisions.
- AI should assist—not replace—human judgment.
- Mechanisms must allow human intervention or override.
- Ethical decision-making must always involve human responsibility.
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8) Sustainability

- AI should promote environmental and social sustainability.
- Energy-efficient algorithms and infrastructure should be encouraged.
- AI must contribute to solving global challenges like climate change.
- Long-term societal benefits must be prioritized over short-term gains.

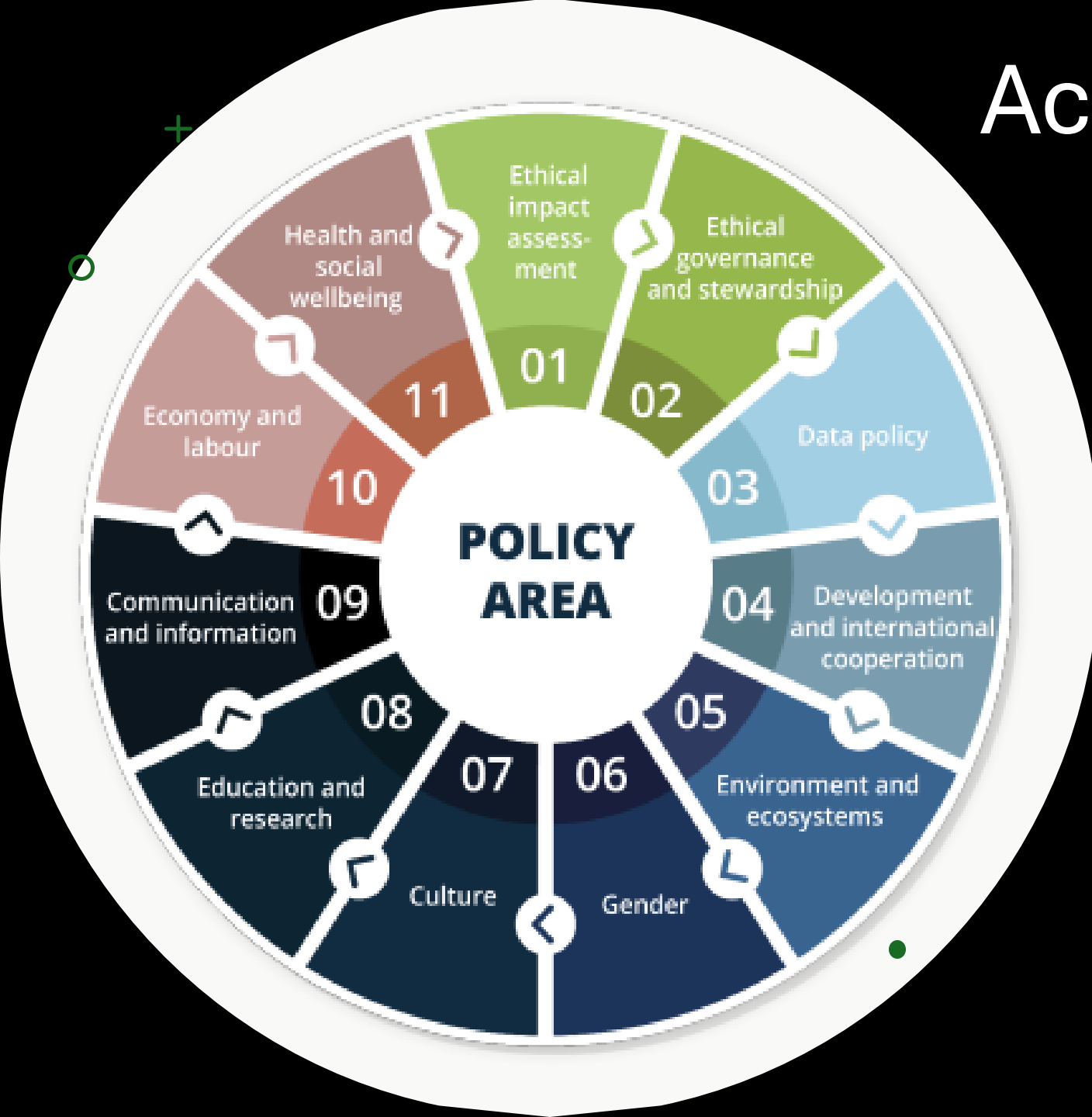
9) Awareness and Literacy

- Public awareness about AI benefits and risks must be increased.
- AI education should be included in academic curricula.
- Professionals must be trained in ethical AI practices.
- Informed citizens can make responsible decisions about AI use.
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10) Fairness and Non-Discrimination

- AI must avoid bias based on gender, race, or socioeconomic status.
- Diverse datasets should be used to reduce algorithmic bias.
- AI systems must promote equal opportunities and inclusion.
- Continuous monitoring is required to detect and mitigate unfair outcomes.

Actionable Policies



IMPLEMENTING THE RECOMMENDATION

- 1. READINESS ASSESSMENT METHODOLOGY (RAM)
- Description: The RAM is designed to help assess whether Member States are prepared to effectively implement the Recommendation.
- The RAM may address questions such as:
 - Are laws in place to regulate AI?
 - Does the national infrastructure support the accessibility of AI technologies?

Purpose: This methodology will help Member States identify their status of preparedness and provide a basis for UNESCO to custom-tailor its capacity-building support. It may also be used to help Member States develop a roadmap towards ethical development and use of AI

- 2. ETHICAL IMPACT ASSESSMENT (EIA)
- Description: EIA is a structured process which operationalises the Recommendation by helping AI project teams, in collaboration with the affected communities, to identify and assess the impacts an AI system may have.
- The EIA may address questions such as:
 - Who is most likely to be adversely affected by this AI system?
 - What form will these impacts take?
 - What can be done to prevent these harms and have resources been allocated to this harm prevention?

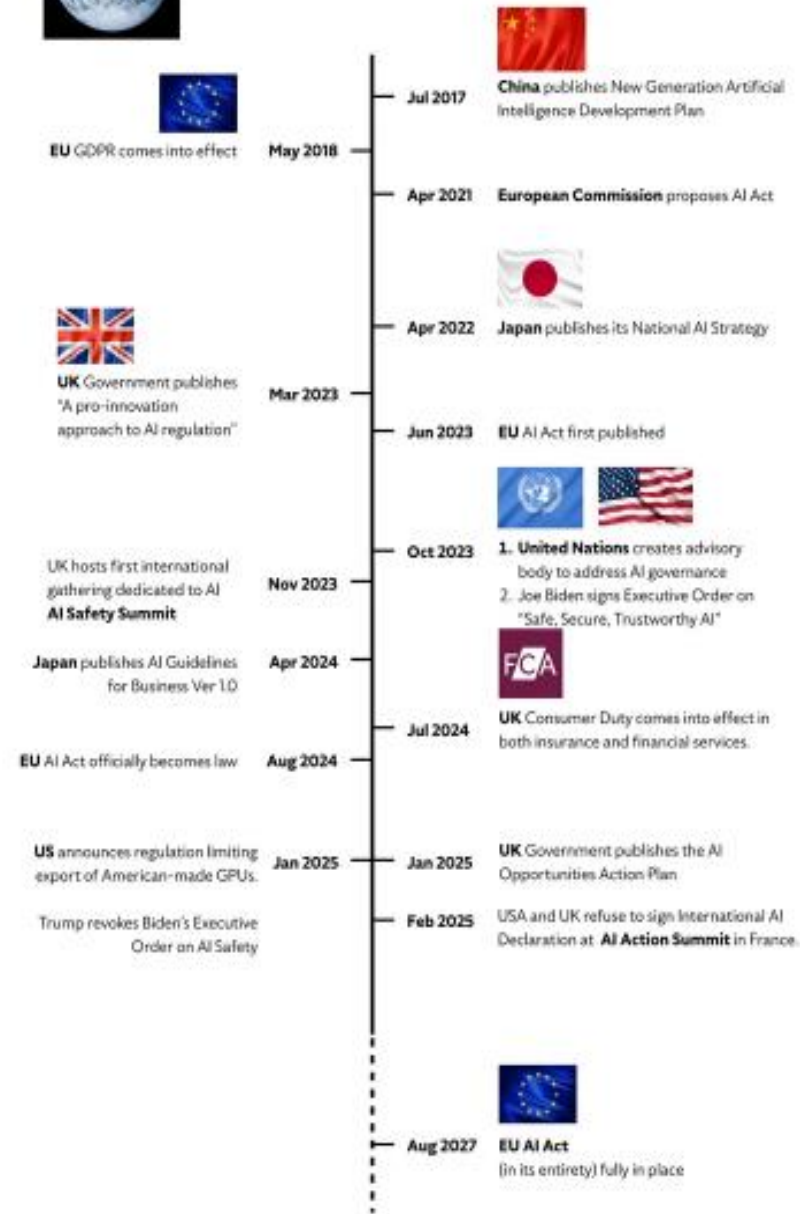
Purpose: EIA provides an opportunity to reflect on the potential impacts of an AI project and to identify the needed harm prevention actions.

AI Laws and Policies: International and regional perspectives.

- Australia- [National AI Plan](#)
- India- [National Strategy for AI](#), [India AI Governance Guidelines](#)
- In the UAE, President Sheikh Mohamed Bin Zayed Al Nahyan [announced the creation of the Artificial Intelligence and Advanced Technology Council \(AIATC\)](#) in January 2024.
- In Saudi Arabia, the first regulatory framework in relation to AI, the [AI Ethics Principles](#), was published in 2023. Saudi Arabia is actively advancing its AI and data-driven innovation as part of its [Vision 2030](#) economic diversification strategy.



Timeline of AI Regulations around the world



AI Regulations around the World - 2026

Intellectual Property and AI-Generated Content

- The Legal Framework: Indian Copyright Act, 1957
- **The Copyright Act, 1957** is the primary legislation governing copyright protection in India. It recognises authorship as the foundation of copyright and grants exclusive rights to authors for their original works. However, the Act does not expressly define or include AI systems as authors.
- **Authorship and Originality: Section 17 of the Act** states that the author of a work is the person who creates it. The term ‘person’ is generally understood to mean a natural human being, not a machine or software.
- **Computer-Generated Works:** The Act was amended in 1994 to include Section 2(d)(vi), which defines “author” for computer-generated works as the person who causes the work to be created. This provision recognises that a computer may generate works autonomously but places authorship with the human agent responsible for such creation.
- **Registration Requirements:** Application forms for copyright registration (Form XIV) require the claimant’s details such as name, nationality, and address, which presupposes human authorship.

Who Can Be Considered the Author of AI-Generated Works?

- There are three potential candidates who could claim authorship over AI-generated content in India:
- **AI Systems:** Currently, AI programs and systems are not recognised as legal persons under Indian law. They cannot own property, including copyrights. Consequently, AI itself cannot be an author or hold copyright.
- **AI Developers:** Ownership claims by developers depend on the terms of use and contractual agreements. Most AI platforms like ChatGPT explicitly disclaim ownership over user-generated content. Moreover, just as Microsoft is not an author of a drawing made using Microsoft Paint, AI developers typically are not considered authors of the content generated by users employing their software.
- **Users/Prompt-Givers:** **Section 2(d)(vi)** suggests that the person who causes the computer to create a work is the author. However, simply entering a prompt or command may not amount to sufficient human creativity or skill to qualify as authorship.

The Essential Elements for Copyright Protection

- In India, for a work to be copyrightable, two core requirements must be met:
- **Originality:** The work must be original in the sense of reflecting the author's skill, judgement, and effort. The Supreme Court in *Eastern Book Company v. D.B. Modak* clarified that originality means a product of skill and judgement that is not purely mechanical or trivial. Mere “sweat of the brow” (effort alone) is insufficient, and the U.S. standard of “modicum of creativity” is deemed too high for Indian law.
- **Human Authorship:** Copyright law aims to incentivise human creativity. The Constitution's Copyright Clause reflects this by granting rights to human authors and inventors. Courts have consistently upheld that human intervention and creativity are necessary for copyright protection.

Challenges Posed by AI-Generated Works

- AI systems generate content by processing vast amounts of existing data. This raises two key challenges:
- **Originality Question:** Since AI outputs often recombine pre-existing material, are they truly original? Indian courts require that originality comes from human skill and judgement, which may be lacking if AI generates works independently.
- **Authorship and Ownership:** When AI acts autonomously or with minimal human input, it becomes difficult to attribute authorship to a human, raising questions about ownership and enforceability of rights.

The ‘Significant Human Input’ Test: A Practical Approach

- To bridge the gap between technology and law, a “Significant Human Input” test is increasingly suggested by legal scholars and practitioners. This approach involves two prongs:
- a) Originality through Human Skill and Judgement: The human author’s contribution must go beyond mere prompting or trivial input. It should involve meaningful creative decisions, such as selecting, editing, rearranging, or reshaping AI outputs.
- b) Quantitative and Qualitative Extent of Human Involvement: The degree of human input should be substantial enough that the final work either would not exist or would be fundamentally different without it.
- This test respects the traditional copyright framework by insisting on human creativity, while recognising that AI can be a powerful tool in the creative process.

Illustrative Examples

- Consider these scenarios:
- Pure AI Generation: An AI writes a story in response to a simple prompt, and the user publishes it without changes. This work lacks significant human input and is unlikely to be protected under copyright law in India.
- Human Editing and Rewriting: A user obtains an AI-generated draft and extensively revises plot, characters, language, and themes. Here, the human involvement crosses the threshold of significant input, and copyright protection may be granted to the human author.
- AI-Assisted Form Corrections: An author writes an article and uses AI tools to correct grammar and improve style. Since the core content is the human's creation, copyright protection clearly applies.
- Research Enhancement: A legal scholar uses AI to summarise judgments and identify key points but performs original analysis and drafting. The human creativity is the dominant factor, making the work copyrightable.

Fair Use Considerations

- Indian copyright law allows limited use of copyrighted material without permission under the “fair use” exceptions for purposes such as criticism, research, or education. While AI training datasets often include copyrighted works, whether such use qualifies as fair use remains an unsettled question. Similarly, whether AI-generated outputs themselves benefit from fair use depends on their nature, purpose, and market impact, requiring case-by-case analysis.

Judicial and Policy Perspectives

- Indian courts have yet to directly rule on the copyrightability of purely AI-generated works. However, decisions like *Eastern Book Company and Navigators Logistics Ltd. v. Kashif Qureshi* emphasise the need for human involvement. Internationally, courts and copyright offices are gradually evolving frameworks to accommodate AI’s role, often reinforcing the primacy of human creativity.
- From a policy standpoint, extending copyright protection only to works with significant human input discourages “free riding” on AI-generated content and supports fair remuneration for original human creators. At the same time, it encourages the use of AI as a creative aid rather than a replacement.
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Conclusion

- AI-generated works are challenging the very foundations of copyright law, which has traditionally protected human creativity and originality. In India, the absence of explicit statutory guidance means that existing principles continue to govern.
- The key takeaway is that copyright protection will remain reserved for works where significant human creativity and effort shape the final output, even if AI plays a role in the process.
- The “Significant Human Input” test strikes a balance between embracing the benefits of AI in creative industries and preserving the core purpose of copyright to incentivise human innovation.

Future of AI Ethics

Some of the future directions of AI ethics are:

- **Automated Auditing Tools:** AI-driven evaluations will verify model fairness, privacy and safety before deployment.
- **Global Standardization:** Unified guidelines will emerge to ensure consistent ethical behavior across industries.
- **Better Explainability Models:** Research will improve interpretability, making deep neural decisions easier to understand.
- **Ethical Certifications:** Products may require compliance badges similar to quality standards used in manufacturing.
- **Continuous Monitoring Systems:** Real-time feedback loops will detect ethical violations as user behavior evolves.
- **Human-AI Collaboration:** Ethical frameworks will encourage co-decision models, blending human judgment with automated insights.

- The future of AI ethics is shaped by the need to address emerging issues and develop effective governance frameworks. Key directions include:
- **Proactive Governance:** Establishing clear guidelines and regulations to ensure AI technologies align with human values and societal well-being.
- **Ethical AI Design:** Incorporating ethical principles from the design stage to prevent bias and ensure transparency.
- **Global Cooperation:** Fostering collaboration among nations to address AI-related challenges and promote ethical AI development.
- **Ethics-by-Design:** Integrating ethical considerations into the development process of AI systems to prevent unintended consequences.
- **Societal Considerations:** Evaluating how AI impacts jobs, culture, and human rights to ensure responsible adoption at scale.
- These directions aim to mitigate risks and ensure that AI technologies are developed and used in a way that respects human dignity and promotes societal welfare.

AI in healthcare

- What Is AI in Healthcare?
- AI in healthcare refers to the use of machine learning, natural language processing, deep learning and other AI technologies to enhance the experiences of both healthcare professionals and patients.
- The data-processing and predictive capabilities of AI enable health professionals to better manage their resources and take a more proactive approach to various aspects of healthcare.
- With these technologies, doctors can then make quicker and more accurate diagnoses, health administrators can locate electronic health records faster and patients can receive more timely and personalized treatments.

AI in Healthcare

- Artificial intelligence simplifies the lives of patients, doctors and hospital administrators by performing tasks that are typically done by humans, but in less time and at a fraction of the cost.
- ChatGPT Health and Claude for Life Sciences are enabling users to connect their health data from wearable devices directly into AI models for hyperpersonalized recommendations and to predict potential health risks.

Examples of AI in Healthcare

- AI in Personal Health
- AI in Medical Diagnosis
- AI in Drug Discovery
- AI in Patient Experience
- AI in Healthcare Data Management
- AI in Robotic Surgery

Uses for AI in Healthcare

- Improving medical diagnosis
- Speeding up drug discovery
- Transforming patient experience
- Managing healthcare data
- Performing robotic surgery

AI in Education

- Artificial intelligence is transforming education by personalizing learning, supporting teachers, and expanding access, while raising questions about equity, integrity, and pedagogy.
- **Key Applications of AI in Education**
- AI in education refers to digital tools that use algorithms and predictive modeling to assist with learning, planning, assessment, and instruction. Common applications include:

Discovery Education

- Adaptive Learning Platforms: AI adjusts content difficulty and provides personalized feedback based on student performance, helping learners progress at their own pace.
- AI Tutors and Assistants: Chat-based AI can explain concepts, provide practice exercises, and offer one-on-one support, sometimes in local languages or low-bandwidth settings.
- Administrative Support: AI automates routine tasks such as grading, lesson planning, and identifying patterns in student data, freeing teachers to focus on instruction and student engagement.
- Generative AI Tools: High school and college students increasingly use tools like ChatGPT for brainstorming, essay editing, research, and coding assistance, enhancing creativity and efficiency.
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Benefits

- AI can improve educational outcomes by personalizing learning, providing immediate feedback, and supporting teachers in managing workloads.
- It can also expand access to quality education, particularly in underserved areas, and reduce stress for students by offering guidance and support.
- Schools report that AI helps reclaim time for teachers, allowing them to focus on high-value teaching and student relationships.

Challenges and Risks

Despite its benefits, AI introduces several challenges:

- **Equity and Access:** Unequal access to AI tools can deepen educational disparities.
- **Academic Integrity:** Generative AI raises concerns about plagiarism and the authenticity of student work.
- **Pedagogical Uncertainty:** Teachers and institutions are still determining the most effective ways to integrate AI without undermining critical thinking or learning skills.
- **Privacy and Dependency:** AI companions and chatbots may affect student privacy and foster over-reliance on technology for social or emotional support.

AI in Governance

- Artificial intelligence (AI) governance refers to the processes, standards and guardrails that help ensure AI systems and tools are safe and ethical.
- AI governance frameworks direct AI research, development and application to help ensure safety, fairness governance refers to the processes, standards and guardrails that help ensure AI systems and tools are safe and ethical.
- <https://www.ibm.com/think/topics/ai-governance>